

luwa-ul Package Documentation

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1 設置 (Installation)

インストール方法は、適切な箇所に `luwa-ul.sty` を配置し、`\usepackage{luwa-ul}` と書き込むだけで完了です。オプションはありません。(Place `luwa-ul.sty` in a directory where LaTeX can find it.)

依存性についてです。(Dependencies:)

- `luacolor`, `xcolor`
- `lua-ul`
- `calc`, `tikz`
- `luatexja-adjust` (from v1.2.5 onwards)

を内部的に仕様しています。(This package uses them internally.)

2 注意 (Caution)

このパッケージはその実装の上で Lua 言語を仕様しているので、LuaLaTeX の上でのみ仕様できるパッケージとなっています。(This package internally uses Lua, so it can be compiled only by LuaLaTeX.)

3 提供されるコマンド (Commands)

提供されるコマンドは以下の通りです。要望に応じて追加されることもありえます。(The provided commands are as follows. I would add some more commands upon requests.)

1.Manual

- `\underLineKK`, `\overLineKK`
- `\niyyusen`, `\keshiniyyusen`
- `\thinHighLight`, `\thickHighLight`
- `\tenKK`, `\dashKK`, `\namiKK`
- `\LineNumbering`

2.Auto

- `\underLineKKAUTO`, `\overLineKKAUTO`
- `\niyyusenAUTO`
- `\thinHighLightAUTO`, `\thickHighLightAUTO`
- `\tenKKAUTO`, `\dashKKAUTO`, `\namiKKAUTO`

使い分けは、線を突き破り得る引数 (ルビやインテグラルなど) をとる場合には Auto シリーズを推奨、それ以外の用途では Manual シリーズを推奨としておきます。ただし、縦書きでは基本的に Auto

シリーズをメインとすることを推奨とします。

3.1 underLineKK

このコマンドは、lua-ul パッケージによって提供される `\underLine` コマンドに対し、

- 縦書き時のずれの解消
- ルビがあった際など、高さや深さが通常の文字と変わる場合に対する挙動の変化

の2点において調整を施したものとなっています。

(This command is basically made out of `\underLine` command, which is provided by `lua-ul` package. I added some adjustments on it in order to

- eliminate the misalignment in vertical writing mode.
- change its behavior when the argument includes taller or deeper than normal characters.

)

使い方は以下のようになります。(You can use it as follows:)

Input

```
1 % 構文
2 \underLineKK[color=<色>, bottom=<下線最下部の位置のずれ>,
3 height=<下線の太さ (bottomの位置からの距離で計算) >]{<テキスト>}
4
5 % 使用例
6 \underLineKK[color=red]{ああああああああ}
7 \underLineKK[bottom=.5\zw]{ああああああああ}
8 \underLineKKAuto{ああ\ruby{あ}{あ}ああああ}
9
10 \parbox<t>[c][6cm][c]{3cm}{%
11 \underLineKK[color=red]{ああああああああ}
12 \underLineKK[bottom=.5\zw]{ああああああああ}
13 \underLineKKAuto{ああ\ruby{あ}{あ}ああああ}
14 }
```

Output

横書き：

ああああああああああああああああああああああ^あ

縦書き：

あ	あ	あ
あ	あ	あ
あ	あ	あ
あ	あ	あ
あ	あ	あ
あ	あ	あ
あ	あ	あ
あ	あ	あ
あ	あ	あ
あ	あ	あ

3.2 \overLineKK

このコマンドは`\underLineKK` を使用しても作成が可能ですが、ショートカットとして用意しておきました。このコマンドは縦書き環境においては通常の`\underLineKK` につけてあったルビ補正を抜いてあります。(This command can also be created using `\underLineKK`, but it has been provided as a shortcut. In a vertical writing environment, this command omits the ruby adjustment that was included in the standard `\underLineKK`.)

Input

```

1  % 構文
2  \overLineKK[color=<色>]{<テキスト>}
3
4  %使用例
5  \overLineKK[color=red]{ああああああああ}
6  \overLineKK{ああ\ruby{あ}{あ}ああああ}
7  \overLineKKAUTO{ああ\ruby{あ}{あ}ああああ}
8
9  \parbox<t>[c][6cm][c]{3cm}{%
10 \overLineKK[color=red]{ああああああああ}
11 \overLineKK{ああ\ruby{あ}{あ}ああああ}
12 \overLineKKAUTO{ああ\ruby{あ}{あ}ああああ}
13 }

```

Output

横書き：

ああああああああああああああああああああああああああ

縦書き：

あ あ あ あ あ
あ あ あ あ あ
あ あ あ あ あ

3.3 \nijyusen, \keshinijyusen

これらのコマンドは`\underlineKK`を入れ子にすることでも作成が可能です、`keshinijyusen` に関してはルビの有無による補正をなくしてあります。(These commands can also be created by nesting `\underlineKK`, but for `keshinijyusen`, the adjustment for the presence or absence of ruby has been removed.)

```

1 % 構文
2 \niyusen[color=<色>, bottom=<下線最下部の位置のずれ>]{<テキスト>}
3 \keshiniyusen[color=<色>]{<テキスト>}
4
5
6 % 使用例
7 \niyusenAuto[color=red]{ああ\ruby{あ}{あ}あああああ}
8 \keshiniyusen{ああ\ruby{あ}{あ}あああああ}
9 \niyusen[color=red]{あああああああ}
10 \keshiniyusen[color=blue]{あああああああ}
11
12 \parbox<t>[c][6cm][c]{3cm}{%
13 \niyusenAuto[color=red]{ああ\ruby{あ}{あ}あああああ}
14 \keshiniyusen{ああ\ruby{あ}{あ}あああああ}
15 \niyusen[color=red]{あああああああ}
16 \keshiniyusen[color=blue]{あああああああ}
17 }

```

横書き：

あ あ

縦書き：

あ あ あ あ あ あ あ あ あ あ あ あ
あ あ あ あ あ あ あ あ あ あ あ あ
あ あ あ あ あ あ あ あ あ あ あ あ

3.4 \thinHighLight, \thickHighLight

これらは蛍光ペンを引いたような出力を提供するコマンドです。ルビの有無に対する挙動が `\underlineKK` のそれとは少々異なり、単なる並行移動だけではなく線の太さ自体も追隨して動くという仕様にしています。(These are commands that provide an output resembling a highlight made with a fluorescent marker. Their behavior regarding the presence or absence of ruby differs slightly from that of `\underlineKK`, as they are designed not only to shift position but also to have the line thickness itself adjust accordingly.)

```

1 % 構文
2 \thinHighLight [color=<色>]{<テキスト>}
3 \thickHighLight [color=<色>]{<テキスト>}
4
5 % 使用例
6 \thinHighLightAuto{あああ\ruby{あ}{あ}ああああ}
7 \thickHighLightAuto [color=green]{ああ\ruby{あ}{あ}あああああ}
8 \thinHighLight{ああああああああ}
9 \thickHighLight [color=green]{あああああああああ}
10 \thinHighLightAuto{あああ\ruby{あ}{あ}ああああ}
11 \thickHighLightAuto{ああ$\displaystyle\int_{b}^{a}x^2\, dx$ああああ}
12 \thinHighLightAuto{ああ$\displaystyle\int_{b}^{a}x^2\, dx$ああああ}
13
14 \parbox<t>[c][6cm][c]{3cm}{%
15   \thinHighLightAuto{あああ\ruby{あ}{あ}ああああ}
16   \thickHighLightAuto [color=green]{ああ\ruby{あ}{あ}あああああ}

```

```

17 \thinHighLight{あああああああ}
18 \thickHighLight[color=green]{あああああああ}
19 \thickHighLightAuto{ああ$\displaystyle\int_b^ax^2\, dx$あああ}
20 \thinHighLightAuto{ああ$\displaystyle\int_b^ax^2\, dx$あああ}
21 }

```

Output

横書き：

あああああああ あああああああ ああああああああ あああああああああ ああああああああ
ああ $\int_b^a x^2 dx$ あああああ ああ $\int_b^a x^2 dx$ あああああ

縦書き：

ああ $\int_b^a x^2 dx$ あああああ	ああ $\int_b^a x^2 dx$ あああああ	ああ $\int_b^a x^2 dx$ あああああ	ああ $\int_b^a x^2 dx$ あああああ
----------------------------	----------------------------	----------------------------	----------------------------

3.5 \namiKK

このコマンドは波線を引くコマンドを提供します。ルビに対する挙動は`\underLineKK` のそれに同じです。(This command provides a wavy underline. Its behavior with respect to ruby is the same as that of `\underLineKK`.)

Input

```

1 % 構文
2 \namiKK[color=色, bottom=下線最下部の位置のずれ]{引数}
3
4 % 使用例
5 \namiKK[color=green]{あああ\ruby{あ}{あ}あああ}
6 \namiKKAUTO{あああ\ruby{あ}{あ}あああ}
7 \namiKK[bottom=-.5\zw]{あああああああああ}
8
9 \parbox<t>[c][6cm][c]{3cm}{%
10 \namiKK[color=green]{あああ\ruby{あ}{あ}あああ}
11 \namiKKAUTO{あああ\ruby{あ}{あ}あああ}
12 \namiKK[bottom=-.5\zw]{あああああああああ}

```


Output

横書き：

ああああああああああああああああああああああ

縦書き：

ああああ
ああああ
ああああ
ああああ
ああああ
ああああ
ああああ
ああああ
ああああ
ああああ

3.6 \tenKK, \dashKK

これらはそれぞれ、点線、破線を提供します。ルビに対する挙動は\underLineKK のそれと同じです。(These commands provide dotted and dashed underlines, respectively. Their behavior toward ruby text is the same as that of \underLineKK.)

Input

```

1 \tenKK[color=色, bottom=下線最下部の位置のずれ, radius=ドットの半径, distance=
  ドット間距離]{引数}
2 % デフォルト : radius=.05\zw, distance=.2\zw
3 \dashKK[color=色, bottom=下線最下部の位置のずれ, width=破線の太さ, distance=破線
  間距離, length=破線の長さ]{引数}
4 % デフォルト : width=.2pt, length=.2\zw, distance=.15\zw
5
6 \tenKKAuto[color=blue]{ああああ\ruby{あ}{あ}ああ}
7 \tenKK[radius=\zw, distance=.4\zw]{ああああああああ}
8 \dashKKAuto[color=red]{ああああ\ruby{あ}{あ}ああ}
9 \dashKK[length=.05\zw, distance=.3\zw]{ああああああああ}
10
11 \parbox<t>[c][6cm][c]{3cm}{%
12   \tenKKAuto[color=blue]{ああああ\ruby{あ}{あ}ああ}
13   \tenKK[radius=\zw, distance=.4\zw]{ああああああああ}
14   \dashKKAuto[color=red]{ああああ\ruby{あ}{あ}ああ}
15   \dashKK[length=.05\zw, distance=.3\zw]{ああああああああ}

```

Output

横書き：

Horizontal output of the character 'あ'. The text 'あ' is repeated. The first 'あ' has a blue underline. The second 'あ' has a red underline. The third 'あ' has a blue underline. The fourth 'あ' has a red underline. The fifth 'あ' has a blue underline. The sixth 'あ' has a red underline. The seventh 'あ' has a blue underline. The eighth 'あ' has a red underline. The ninth 'あ' has a blue underline. The tenth 'あ' has a red underline. The eleventh 'あ' has a blue underline. The twelfth 'あ' has a red underline. The thirteenth 'あ' has a blue underline. The fourteenth 'あ' has a red underline. The fifteenth 'あ' has a blue underline. The sixteenth 'あ' has a red underline. The seventeenth 'あ' has a blue underline. The eighteenth 'あ' has a red underline. The nineteenth 'あ' has a blue underline. The twentieth 'あ' has a red underline. The twenty-first 'あ' has a blue underline. The twenty-second 'あ' has a red underline. The twenty-third 'あ' has a blue underline. The twenty-fourth 'あ' has a red underline. The twenty-fifth 'あ' has a blue underline. The twenty-sixth 'あ' has a red underline. The twenty-seventh 'あ' has a blue underline. The twenty-eighth 'あ' has a red underline. The twenty-ninth 'あ' has a blue underline. The thirtieth 'あ' has a red underline. The thirty-first 'あ' has a blue underline. The thirty-second 'あ' has a red underline. The thirty-third 'あ' has a blue underline. The thirty-fourth 'あ' has a red underline. The thirty-fifth 'あ' has a blue underline. The thirty-sixth 'あ' has a red underline. The thirty-seventh 'あ' has a blue underline. The thirty-eighth 'あ' has a red underline. The thirty-ninth 'あ' has a blue underline. The fortieth 'あ' has a red underline. The forty-first 'あ' has a blue underline. The forty-second 'あ' has a red underline. The forty-third 'あ' has a blue underline. The forty-fourth 'あ' has a red underline. The forty-fifth 'あ' has a blue underline. The forty-sixth 'あ' has a red underline. The forty-seventh 'あ' has a blue underline. The forty-eighth 'あ' has a red underline. The forty-ninth 'あ' has a blue underline. The fiftieth 'あ' has a red underline. The fifty-first 'あ' has a blue underline. The fifty-second 'あ' has a red underline. The fifty-third 'あ' has a blue underline. The fifty-fourth 'あ' has a red underline. The fifty-fifth 'あ' has a blue underline. The fifty-sixth 'あ' has a red underline. The fifty-seventh 'あ' has a blue underline. The fifty-eighth 'あ' has a red underline. The fifty-ninth 'あ' has a blue underline. The sixtieth 'あ' has a red underline. The sixty-first 'あ' has a blue underline. The sixty-second 'あ' has a red underline. The sixty-third 'あ' has a blue underline. The sixty-fourth 'あ' has a red underline. The sixty-fifth 'あ' has a blue underline. The sixty-sixth 'あ' has a red underline. The sixty-seventh 'あ' has a blue underline. The sixty-eighth 'あ' has a red underline. The sixty-ninth 'あ' has a blue underline. The seventieth 'あ' has a red underline. The seventy-first 'あ' has a blue underline. The seventy-second 'あ' has a red underline. The seventy-third 'あ' has a blue underline. The seventy-fourth 'あ' has a red underline. The seventy-fifth 'あ' has a blue underline. The seventy-sixth 'あ' has a red underline. The seventy-seventh 'あ' has a blue underline. The seventy-eighth 'あ' has a red underline. The seventy-ninth 'あ' has a blue underline. The eightieth 'あ' has a red underline. The eighty-first 'あ' has a blue underline. The eighty-second 'あ' has a red underline. The eighty-third 'あ' has a blue underline. The eighty-fourth 'あ' has a red underline. The eighty-fifth 'あ' has a blue underline. The eighty-sixth 'あ' has a red underline. The eighty-seventh 'あ' has a blue underline. The eighty-eighth 'あ' has a red underline. The eighty-ninth 'あ' has a blue underline. The ninetieth 'あ' has a red underline. The ninety-first 'あ' has a blue underline. The ninety-second 'あ' has a red underline. The ninety-third 'あ' has a blue underline. The ninety-fourth 'あ' has a red underline. The ninety-fifth 'あ' has a blue underline. The ninety-sixth 'あ' has a red underline. The ninety-seventh 'あ' has a blue underline. The ninety-eighth 'あ' has a red underline. The ninety-ninth 'あ' has a blue underline. The hundredth 'あ' has a red underline.

縦書き：

Vertical output of the character 'あ'. The text 'あ' is repeated. The first 'あ' has a blue underline. The second 'あ' has a red underline. The third 'あ' has a blue underline. The fourth 'あ' has a red underline. The fifth 'あ' has a blue underline. The sixth 'あ' has a red underline. The seventh 'あ' has a blue underline. The eighth 'あ' has a red underline. The ninth 'あ' has a blue underline. The tenth 'あ' has a red underline. The eleventh 'あ' has a blue underline. The twelfth 'あ' has a red underline. The thirteenth 'あ' has a blue underline. The fourteenth 'あ' has a red underline. The fifteenth 'あ' has a blue underline. The sixteenth 'あ' has a red underline. The seventeenth 'あ' has a blue underline. The eighteenth 'あ' has a red underline. The nineteenth 'あ' has a blue underline. The twentieth 'あ' has a red underline. The twenty-first 'あ' has a blue underline. The twenty-second 'あ' has a red underline. The twenty-third 'あ' has a blue underline. The twenty-fourth 'あ' has a red underline. The twenty-fifth 'あ' has a blue underline. The twenty-sixth 'あ' has a red underline. The twenty-seventh 'あ' has a blue underline. The twenty-eighth 'あ' has a red underline. The twenty-ninth 'あ' has a blue underline. The thirtieth 'あ' has a red underline. The thirty-first 'あ' has a blue underline. The thirty-second 'あ' has a red underline. The thirty-third 'あ' has a blue underline. The thirty-fourth 'あ' has a red underline. The thirty-fifth 'あ' has a blue underline. The thirty-sixth 'あ' has a red underline. The thirty-seventh 'あ' has a blue underline. The thirty-eighth 'あ' has a red underline. The thirty-ninth 'あ' has a blue underline. The fortieth 'あ' has a red underline. The forty-first 'あ' has a blue underline. The forty-second 'あ' has a red underline. The forty-third 'あ' has a blue underline. The forty-fourth 'あ' has a red underline. The forty-fifth 'あ' has a blue underline. The forty-sixth 'あ' has a red underline. The forty-seventh 'あ' has a blue underline. The forty-eighth 'あ' has a red underline. The forty-ninth 'あ' has a blue underline. The fiftieth 'あ' has a red underline. The fifty-first 'あ' has a blue underline. The fifty-second 'あ' has a red underline. The fifty-third 'あ' has a blue underline. The fifty-fourth 'あ' has a red underline. The fifty-fifth 'あ' has a blue underline. The fifty-sixth 'あ' has a red underline. The fifty-seventh 'あ' has a blue underline. The fifty-eighth 'あ' has a red underline. The fifty-ninth 'あ' has a blue underline. The sixtieth 'あ' has a red underline. The sixty-first 'あ' has a blue underline. The sixty-second 'あ' has a red underline. The sixty-third 'あ' has a blue underline. The sixty-fourth 'あ' has a red underline. The sixty-fifth 'あ' has a blue underline. The sixty-sixth 'あ' has a red underline. The sixty-seventh 'あ' has a blue underline. The sixty-eighth 'あ' has a red underline. The sixty-ninth 'あ' has a blue underline. The seventieth 'あ' has a red underline. The seventy-first 'あ' has a blue underline. The seventy-second 'あ' has a red underline. The seventy-third 'あ' has a blue underline. The seventy-fourth 'あ' has a red underline. The seventy-fifth 'あ' has a blue underline. The seventy-sixth 'あ' has a red underline. The seventy-seventh 'あ' has a blue underline. The seventy-eighth 'あ' has a red underline. The seventy-ninth 'あ' has a blue underline. The eightieth 'あ' has a red underline. The eighty-first 'あ' has a blue underline. The eighty-second 'あ' has a red underline. The eighty-third 'あ' has a blue underline. The eighty-fourth 'あ' has a red underline. The eighty-fifth 'あ' has a blue underline. The eighty-sixth 'あ' has a red underline. The eighty-seventh 'あ' has a blue underline. The eighty-eighth 'あ' has a red underline. The eighty-ninth 'あ' has a blue underline. The ninetieth 'あ' has a red underline. The ninety-first 'あ' has a blue underline. The ninety-second 'あ' has a red underline. The ninety-third 'あ' has a blue underline. The ninety-fourth 'あ' has a red underline. The ninety-fifth 'あ' has a blue underline. The ninety-sixth 'あ' has a red underline. The ninety-seventh 'あ' has a blue underline. The ninety-eighth 'あ' has a red underline. The ninety-ninth 'あ' has a blue underline. The hundredth 'あ' has a red underline.

3.7 \LineNumbering

下線に番号を振るためのコマンドです。これは私が作成した `KKsymbols` パッケージ (TeX Live に収録済み) との併用をお勧めしています。スターオプションを付与すると、縦書きの時には引数の中身が回転します¹⁾。(This command assigns numbers to underlines. It is recommended to use it together with my `KKsymbols` package (already included in TeX Live). When the starred option is given, the contents of the argument are rotated in vertical writing²⁾.)

Input

```

1 % 横書き
2 \LineNumbering<スターオプション>{<ラベル>}[<上昇させる距離>]
3
4 % 横書き
5 \LineNumbering{\kakko{1}}\underLineKK{あああああああ}
6
7 % 縦書き
8 \parbox<t>[c][6cm][c]{3cm}{%
9   \LineNumbering{\kakko{1}}\underLineKK{あああああああ}
10  \LineNumbering*{ア}\underLineKK{あああああああ}
11 }
```

1) 横書きの時には出力に変化はありません。

2) In horizontal writing, the output remains unchanged.

$$\int_a^b f(x) dx \quad \int_a^b f(x) dx$$

$$\int_a^b f(x) dx$$

$$\int_a^b f(x) dx$$

$$\int_a^b f(x) dx$$

$$\int_a^b f(x) dx$$

縦書き：

$$\int_a^b f(x) dx \quad \int_a^b f(x) dx \quad \int_a^b f(x) dx \quad \int_a^b f(x) dx \quad \int_a^b f(x) dx \quad \int_a^b f(x) dx$$

以上の比較を参照すると一目瞭然でしょう。

使い分けについてですが、基本的には Manual Series を使用してください。なぜなら Auto Series は、引数の中に一箇所でも極端に深さをとるコマンドが存在した場合、改行後もずっとその調整のまま線を引き続けてしまいます。したがって、パラグラフの大部分にアンダーラインを入れたいというような場合には不向きです。(Regarding the choice of which series to use, please use the Manual Series by default. This is because the Auto Series continues to draw lines with the same vertical adjustment if there is even a single command within its argument that takes an unusually deep depth. Therefore, it is not suitable for cases where you want to underline a large portion of a paragraph.)

例えば、(For example,)

Input

```

1  以上より、
2  \[f(x)=\nameKKAuto{\frac{x+\frac{a}{b}}{x^4+\frac{c}{d}}}\ \text{……}\kako{
   答}\]
```

```

3  \noindent となります。
```

Output

以上より、

$$f(x) = \frac{x + \frac{a}{b}}{\underline{x^4 + \frac{c}{d}}} \dots\dots(\text{答})$$

となります。

以上のような局面においては、数式に下線が被ると格好が悪いため、Auto Series の使い所であると言えます。(in such situations, it would look awkward if the underline overlapped with mathematical expressions, so this is where the Auto Series becomes useful.)

4 ライセンス (License)

Released under the MIT License.

5 Version History

- **v1.0.1 (2025/10/26)** — Initial public release.
- **v1.1.0 (2025/10/29)** — I added Auto Series, and changed some csname of commands.
- **v1.1.1 (2025/11/10)** — Slightly adjusted the move of Auto Series.
- **v1.2.0 (2025/11/25)** — Get rid of specific adjustment when `\ruby` is used in Manual Series and Auto Series takes over the role. Also, slightly changed internal length adjusting parameter.
- **v1.2.1 (2025/12/28)** — Fixed a problem cased by using `ex` scale. Also, `\LineNumbering` was slightly improved.
- **v1.2.5 (2026/01/30)** — Fllowing two points are changed.
 - Fixed a bug affecting behavior within `\scriptstyle` and `\scriptscriptstyle`.
 - Replaced all instances of `\f@size` and `ex` with `\zw` to better optimize for Japanese type-setting environments.